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DS experiment 3

```
#include <stdio.h>

int arr[100], n;

void inputArray();
void displayArray();
void linearSearch();
void binarySearch();
void insertionSort();
void selectionSort();
void shellSort();

int main()
{
   int choice;

   inputArray();

   do
   {
      printf("\n========== MENU ==========\n");
      printf("1. Display Array\n");
      printf("2. Linear Search\n");
      printf("3. Binary Search\n");
      printf("4. Insertion Sort\n");
      printf("5. Selection Sort\n");
      printf("6. Shell Sort\n");
      printf("7. Exit\n");
      printf("Enter your choice: ");
      scanf("%d", &choice);

      switch(choice)
      {
         case 1:
            displayArray();
            break;

         case 2:
            linearSearch();
```

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            break;

        case 3:
            binarySearch();
            break;

        case 4:
            insertionSort();
            printf("\nArray Sorted using Insertion Sort.\n");
            displayArray();
            break;

        case 5:
            selectionSort();
            printf("\nArray Sorted using Selection Sort.\n");
            displayArray();
            break;

        case 6:
            shellSort();
            printf("\nArray Sorted using Shell Sort.\n");
            displayArray();
            break;

        case 7:
            printf("Program Ended.\n");
            break;

        default:
            printf("Invalid Choice!\n");
        }

    } while(choice != 7);

    return 0;
}

void inputArray()
{
    int i;

    printf("Enter number of elements: ");
    scanf("%d",&n);
```

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    printf("Enter elements:\n");

    for(i=0;i<n;i++)
        scanf("%d",&arr[i]);
}

void displayArray()
{
    int i;

    printf("Array: ");

    for(i=0;i<n;i++)
        printf("%d ",arr[i]);

    printf("\n");
}

void linearSearch()
{
    int key,i;

    printf("Enter element to search: ");
    scanf("%d",&key);

    for(i=0;i<n;i++)
    {
        if(arr[i]==key)
        {
            printf("Element found at position %d\n",i+1);
            return;
        }
    }

    printf("Element not found.\n");
}

void binarySearch()
{
    int key;
    int low=0,high=n-1,mid;

    printf("NOTE: Array must be sorted.\n");
    printf("Enter element to search: ");
```

```
    scanf("%d",&key);

    while(low<=high)
    {
        mid=(low+high)/2;

        if(arr[mid]==key)
        {
            printf("Element found at position %d\n",mid+1);
            return;
        }
        else if(arr[mid]<key)
            low=mid+1;
        else
            high=mid-1;
    }

    printf("Element not found.\n");
}

void insertionSort()
{
    int i,j,key;

    for(i=1;i<n;i++)
    {
        key=arr[i];
        j=i-1;

        while(j>=0 && arr[j]>key)
        {
            arr[j+1]=arr[j];
            j--;
        }

        arr[j+1]=key;
    }
}

void selectionSort()
{
    int i,j,min,temp;

    for(i=0;i<n-1;i++)
```

```
    {
        min=i;

        for(j=i+1;j<n;j++)
        {
            if(arr[j]<arr[min])
                min=j;
        }

        temp=arr[i];
        arr[i]=arr[min];
        arr[min]=temp;
    }
}

void shellSort()
{
    int gap,i,j,temp;

    for(gap=n/2;gap>0;gap=gap/2)
    {
        for(i=gap;i<n;i++)
        {
            temp=arr[i];

            for(j=i;j>=gap && arr[j-gap]>temp;j-=gap)
            {
                arr[j]=arr[j-gap];
            }

            arr[j]=temp;
        }
    }
}
```

Output:
Enter number of elements: 4
Enter elements:
56
34
789
67

```
========== MENU ==========
1. Display Array
2. Linear Search
3. Binary Search
4. Insertion Sort
5. Selection Sort
6. Shell Sort
7. Exit
Enter your choice: 1
Array: 56 34 789 67

========== MENU ==========
1. Display Array
2. Linear Search
3. Binary Search
4. Insertion Sort
5. Selection Sort
6. Shell Sort
7. Exit
Enter your choice: 2
Enter element to search: 789
Element found at position 3

========== MENU ==========
1. Display Array
2. Linear Search
3. Binary Search
4. Insertion Sort
5. Selection Sort
6. Shell Sort
7. Exit
Enter your choice: 4

Array Sorted using Insertion Sort.
Array: 34 56 67 789

========== MENU ==========
1. Display Array
2. Linear Search
3. Binary Search
4. Insertion Sort
5. Selection Sort
6. Shell Sort
7. Exit
```

```
Enter your choice: 5

Array Sorted using Selection Sort.
Array: 34 56 67 789

========== MENU ==========
1. Display Array
2. Linear Search
3. Binary Search
4. Insertion Sort
5. Selection Sort
6. Shell Sort
7. Exit
Enter your choice: 6

Array Sorted using Shell Sort.
Array: 34 56 67 789

========== MENU ==========
1. Display Array
2. Linear Search
3. Binary Search
4. Insertion Sort
5. Selection Sort
6. Shell Sort
7. Exit
Enter your choice: 3
NOTE: Array must be sorted.
Enter element to search: 789
Element found at position 4

========== MENU ==========
1. Display Array
2. Linear Search
3. Binary Search
4. Insertion Sort
5. Selection Sort
6. Shell Sort
7. Exit
Enter your choice: 7
Program Ended.
```